Permeability and Porosity from Sparse Data

Generating rock property ensembles with latent diffusion.

Scope

In the field, saturation and pressure are known only at a few locations, and only at the times a sensor records them. Turning that sparse record into full permeability and porosity fields is an inverse problem, and it is under-determined: many different rock fields can produce the same handful of readings.

A single predicted map hides this. It commits to one answer and says nothing about the others that fit the data equally well. The idea here is to return a *set* of maps instead, all consistent with the measurements, and to read the spread across that set as the uncertainty. The engine that produces the set is a diffusion model.

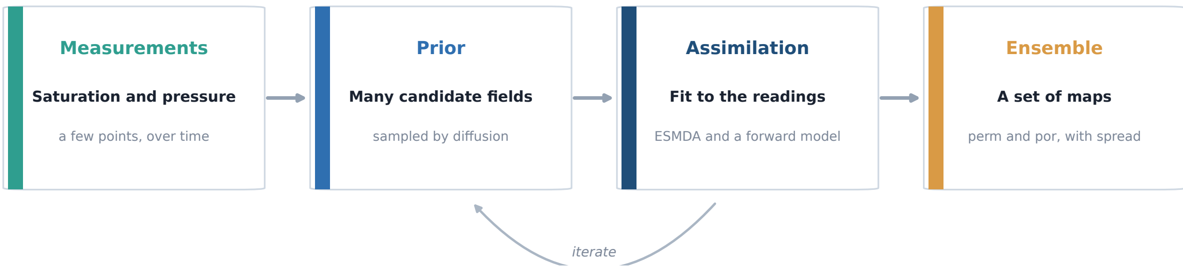

Figure 1. From sparse readings to a set of maps. The prior proposes candidate rock fields, assimilation keeps the ones that reproduce the readings, and the two steps repeat until the set agrees with the data.

Our current GRU-FNO model does not fit this setting. It reads dense saturation and pressure at every cell and every timestep, which is not something a real site provides, and it returns one field with no measure of confidence. The diffusion route relaxes both: sparse input, and a full ensemble as output.

How it works

The method follows recent work on 3D latent diffusion for geomodels (Di Federico and Durlofsky, 2025). Their target is rock facies and ours is continuous permeability and porosity, but the machinery is the same, so we adopt it rather than reinvent it.

Diffusion in one paragraph

A diffusion model is a denoiser. During training it takes a clean field, adds Gaussian noise in small steps until nothing is left, and learns to undo one step of that noise. To generate a new field, it starts from pure noise and denoises its way back to something realistic. Because it starts from noise, a different random start gives a different field. That is exactly what we want: repeat the sampling and a whole ensemble comes out, with no extra machinery.

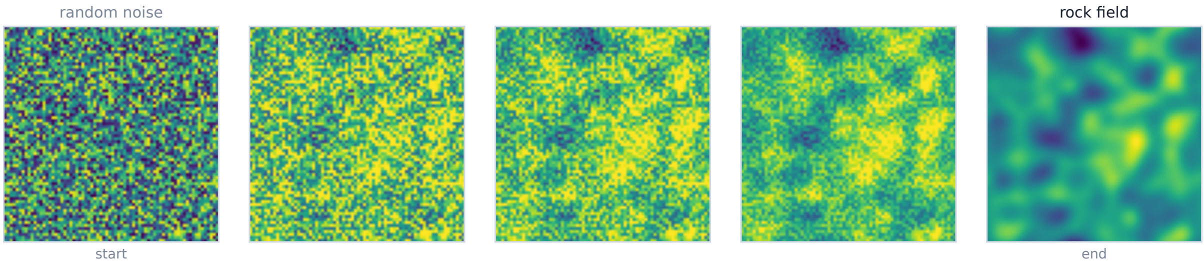

Figure 2. Sampling runs left to right. Structure emerges from noise as the network removes it step by step, ending on a plausible rock field.

Two ingredients that make it practical

Latent space: Running diffusion directly on a full 3D grid is expensive. So a small autoencoder first compresses each field to about a thousand numbers, the diffusion runs in that compact space, and the result is decoded back to full resolution. Same output, far less cost.

Conditioning: The measurements enter through ensemble data assimilation (ESMDA), applied in the compact space. We sample a set of candidate fields, use a fast forward model to predict what each one would read at the sensors, compare that to the real readings, and nudge each candidate so its prediction moves toward the data. Adjusting a thousand numbers per field instead of every grid cell is what keeps this affordable.

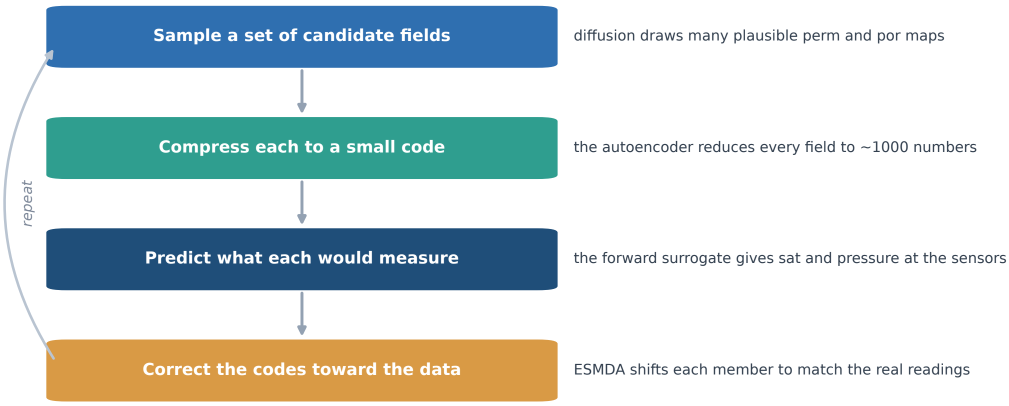

Figure 3. The assimilation loop. Only the small codes are updated, and the loop repeats a few times until the ensemble reproduces the observed readings.

Steps

The build runs in 2D first to keep each part fast to test, then scales to 3D on the larger dataset. The order is deliberate: every step produces something checkable before the next one starts.

1. **Autoencoder in 2D**: Train the compressor on permeability and porosity slices and confirm it reconstructs them cleanly, using the same log and scaling as the current model. If this step is weak, nothing built on top of it can be trusted.

2. **Diffusion in 2D**: Train the latent diffusion and confirm it samples plausible fields on its own, before any measurements are involved. This checks that the prior is sound.

3. **Measurement operator**: Fix the sensor setup: which locations, which timesteps, saturation and pressure. This is the bridge between what a real site can measure and what the model sees.

4. **Assimilation**: Wire ESMDA in the latent space, with the GRU-FNO run forward as the fast forward model, and produce a posterior set for a held-out case.

5. **Validation**: Measure coverage on cases where the truth is known. The question is not only whether the average map is close, but whether the ensemble band actually contains the true field often enough to be trusted. That is the real test of the uncertainty.

6. **Scale to 3D**: Once the 2D recipe holds, repeat it at full grid size on the larger dataset.

The GRU-FNO surrogate is reused as the forward model inside the loop, so the model we already have and the new diffusion prior connect into a single inversion pipeline rather than two separate efforts.

What the result looks like

The point of the whole pipeline is a check we can show, not just a number. A realization is kept only if the signal it produces agrees with the sensors, and that agreement has to hold at the right places and at the right times. The figure below is how we read it: the selected maps on top, and underneath, at each sensor location, their predicted signal tracking the observed readings through time while the rest of the prior drifts away.

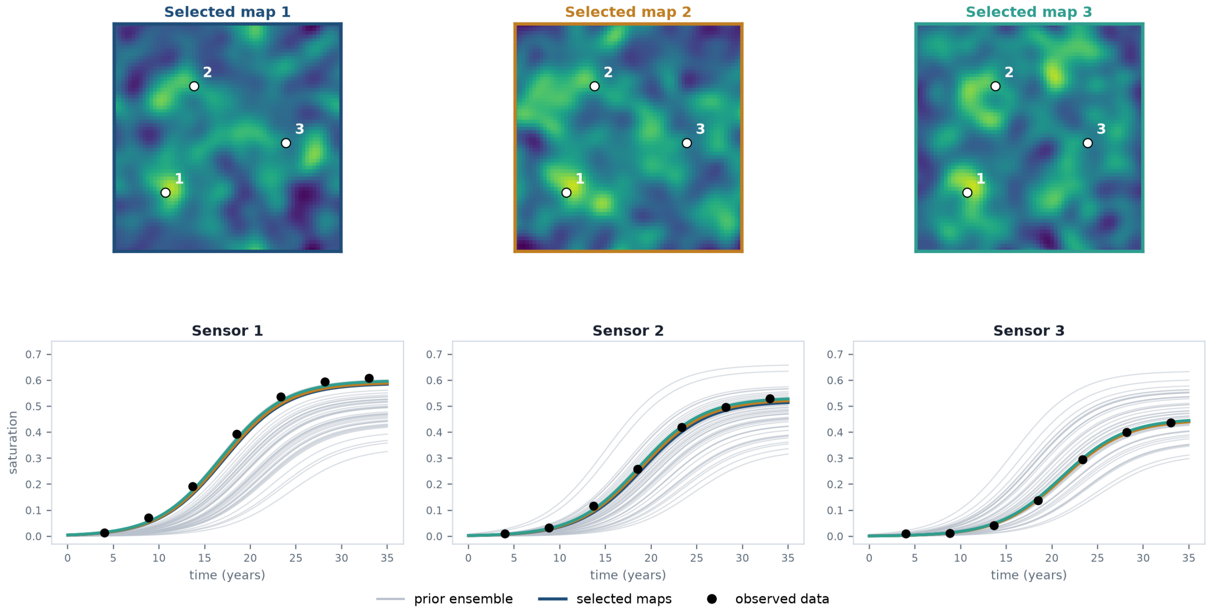

Figure 4. The kept maps match the data at every measurement location and over time. Top: three permeability maps retained from the ensemble, with the sensors marked. Bottom: at each sensor, the signal from those maps (coloured) passes through the observed readings (black dots), while most of the prior ensemble (grey) misses.

The maps and curves here are a synthetic stand-in, drawn to fix the layout before the model exists. Once the diffusion prior and the assimilation loop are trained, the same panels are produced from real fields, and this becomes the standard way we report a case.